AI Agents: Concept, Applications and Evaluations

Leah Lu
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Outline

- AI Agents Intro
- AI Agent Components
- Applications
- Evaluations
- Related Documents
- Q & A

Why We Care About AI Agents?

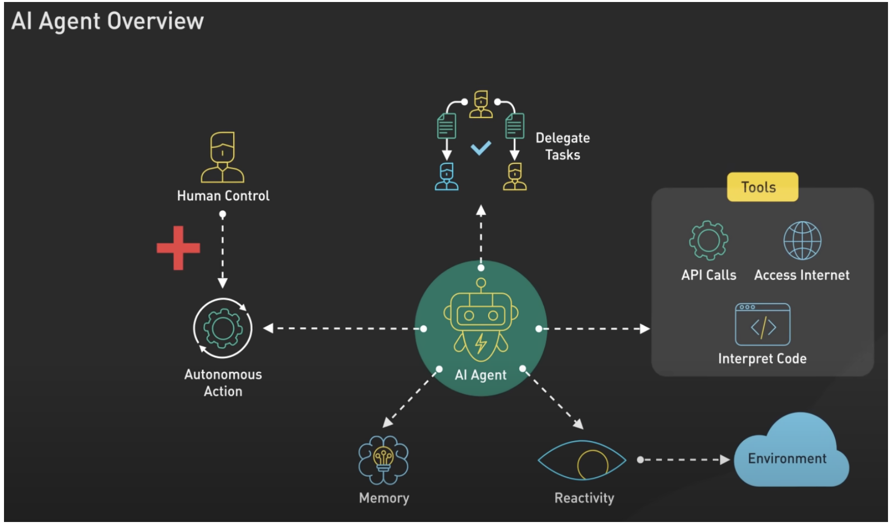
- Word Cloud

What AI Agents or GenAI can do (inspired from GenAI Hackathon)

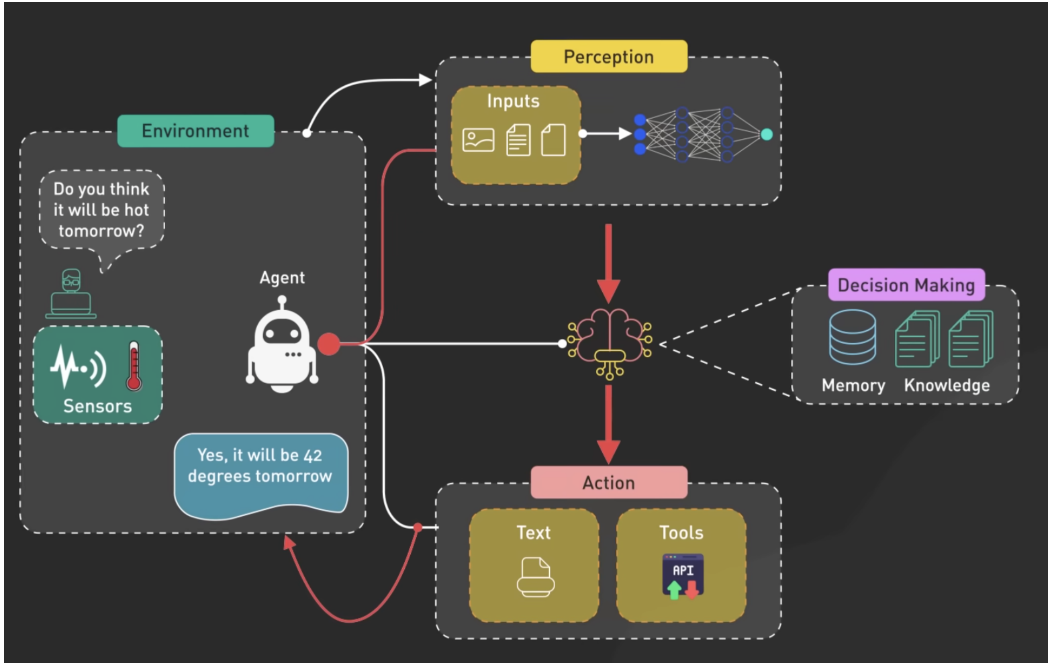
- AI Agents to build a **smart assistant** to enhance data-driven decision-making, planning, and field operations.
- AI Agents to build, accelerate, optimize workflow and understand and **solve problems**
- AI Agents to provide **data analysis** to improve business outcomes by better understanding guests (history, intent, loyalty, needs), products (trends, pricing, forecasts, promotions), and identifying the best time to sell.
- AI Agents (GenAI) to complement our capabilities, including content creation, fixing missing item data, enhancing vendor communication, and optimizing services through **insights**.

What is an AI Agent?

- **Definition for LLM-based agent:** “An autonomous system that can perceive, reason, and act in an environment to achieve goals.”
- **Key Components:** Reasoning and Planning, Memory, Tool Use



AI Agent Example



Assumed processes:

- **Perception**: understand the question
- **Context**: Fill in missing information such as location
- **Decision Making**: whether to use internal knowledge
- **Tool Use**: call weather API
- Generate output

AI Agent Frameworks

LangChain

- Library for AI applications
- Composable components
- Tool and memory integration

LangGraph

- Library for multi-agent flows
- Directed acyclic graph (DAG) structure
- Concurrency management

CREWAI

- Framework for autonomous agents
- Role-based architecture
- Task planning

AutoGen

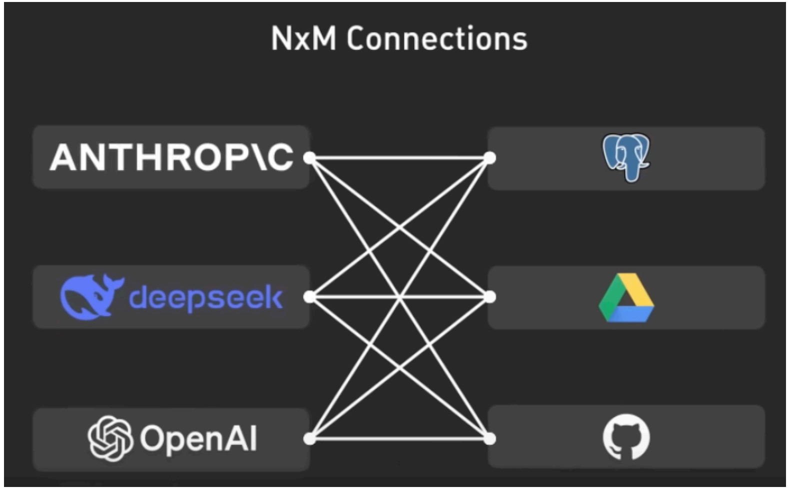
- Framework for LLM applications
- Agent-to-agent conversation
- Scalable customization

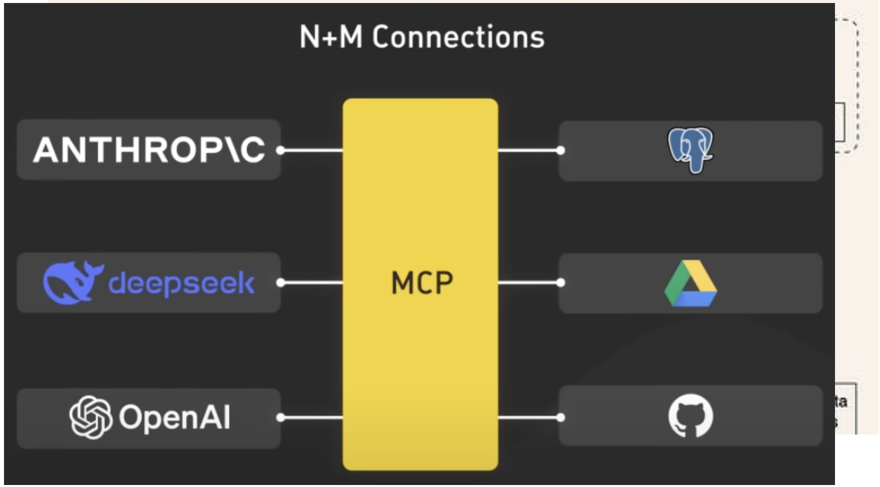

AI Agents for Retail
Personalized Shopping Assistants
Recommend products tailored to customer preferences
Customer Support
Provide assistance and answer customer
Inventory Management
Optimize stock levels and automative
Fraud Detection
Identify and prevent fraudulent transactions
Visual Search & Styling
Enable ima-ge-based product search and styling
Supply Chain
Monitor and optimize logistics and distribution
Logistics Optimization
Improve efficiency in transportation and routing
Merchandising Assistants
Assist in product displays and inventory planning

Challenges for AI Agents

- **Reasoning & Planning:** Struggle with multi-step tasks, goal decomposition, and consistency.
- **Tool Use:** Fragile orchestration, misuse of tools, or misinterpretation of outputs.
- **Memory:** Poor long-term memory and state tracking across steps or sessions.
- **Context Construction:** Difficult to build coherent prompts from multiple inputs (e.g., RAG, goals, history).
- **Evaluation:** Hard to measure success or reliability across diverse tasks.

Model Context Protocol (MCP) and AI Agent





Reference: https://www.youtube.com/watch?v=_d0duu3dED4

AI Agent Components

- Reasoning and Planning, Memory, and Tool Use

Component	With It ✅	Without It ❌
Reasoning	Understands complex goals, makes logical decisions, connects dots	Takes shallow actions, makes irrational or incoherent decisions
Memory	Remembers context, past actions, and user preferences; enables coherent dialogue	Repeats questions, forgets previous steps, lacks continuity
Planning	Breaks tasks into subgoals, sequences steps intelligently	Acts randomly or linearly; gets stuck on complex tasks or misses dependencies
Tool Use	Executes external actions (e.g. API calls, calculations, web search) to augment internal reasoning	Relies only on internal guesses; lacks access to up-to-date or external data

🛒 Use Case: Shopping for a Laptop on an E-Commerce Site

Component	With It ✅	Without It ❌
Reasoning	Understands user needs: “I want a lightweight laptop for travel and coding” → Suggests MacBook Air, Dell XPS	Just recommends random popular laptops without connecting to the user's actual use case
Memory	Remembers preferences: “Last time user liked 13-inch screens & disliked noisy fans” → Filters suggestions	Forgets user’s dislikes and prior purchases; asks the same questions again and again
Planning	Breaks task into steps: “1) Clarify needs → 2) Filter products → 3) Compare → 4) Assist checkout”	Jumps around: shows checkout page before confirming user choice; misses comparison step
Tool Use	Calls external APIs (e.g., real-time price tracker, reviews database, warranty checker) to enrich product suggestions	Lacks access to external data; can't compare prices, verify specs, or check availability

Apple MacBook Air 13
M3 RAM

$1,199.99 reg ~~$1,822.99~~
Sale
Acer Acer Predator Neo 16" Laptop Intel Core i9-14900HX 2.2GHz 16GB RAM 1TB SSD W11H -...
2.0 ★ (1)
Add to cart

How to build the steps for AI agent?

How to guide thinking for the model

ReAct Prompting: Reason and Acting

How to teach model to call the right tool

Toolformer: Language Models Can Teach Themselves to Use Tools

ReAct Prompting: Synergizing Reasoning and Acting in Language Models from Mar. 2023

Query: “*Aside from the Apple Remote, what other devices can control the program?*”

```
(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the
program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in
October 2005 by Apple … originally designed to control the
Front Row media center program …

Thought 2: Apple Remote was originally designed to control
the Front Row media center program. I need to search Front
Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row
Seat to Earth', 'Front Row Motorsports',' Front Row
(software) ', …]

Thought 3: Front Row is not found. I need to search Front
Row (software) .
Act 3: Search[Front Row (software) ]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple
Remote or the keyboard function keys. So the answer is
keyboard function keys.
Act 4: Finish[keyboard function keys ]
```

ReAct operates by guiding the LLM through a cycle of:

1. **Thought**: The model articulates its reasoning about the task at hand.
2. **Action**: Based on its reasoning, the model decides on an action, such as querying a database or searching the web.
3. **Observation**: The model receives the result of its action, which informs its next reasoning step.

This cycle continues until the model arrives at a final answer.

Toolformer: Language Models Can Teach Themselves to Use Tools from Feb. 2023

“We introduce Toolformer, a model trained to decide which APIs to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction.”

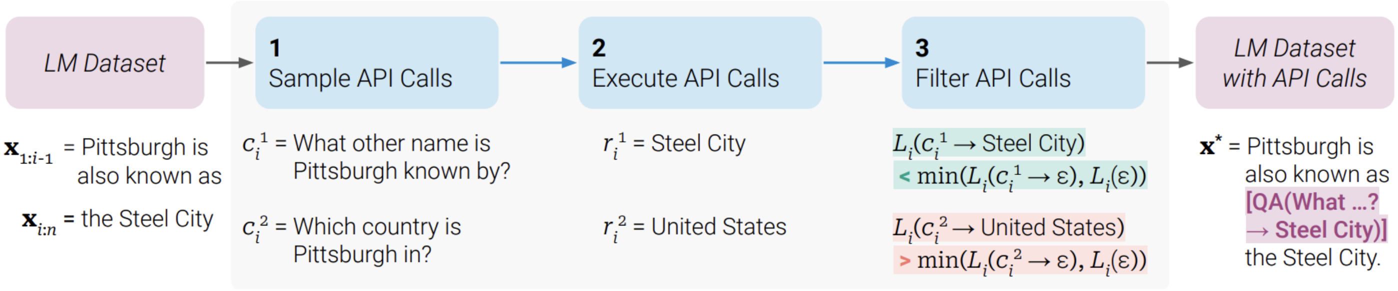


Figure 2: Key steps in our approach, illustrated for a *question answering* tool: Given an input text $\mathbf{x}$, we first sample a position i and corresponding API call candidates $c_i^1, c_i^2, \ldots, c_i^k$. We then execute these API calls and filter out all calls which do not reduce the loss L_i over the next tokens. All remaining API calls are interleaved with the original text, resulting in a new text $\mathbf{x}^*$.

API Name	Example Input	Example Output
Question Answering	Where was the Knights of Columbus founded?	New Haven, Connecticut
Wikipedia Search	Fishing Reel Types	Spin fishing > Spin fishing is distinguished between fly fishing and bait cast fishing by the type of rod and reel used. There are two types of reels used when spin fishing, the open faced reel and the closed faced reel.
Calculator	27 + 4 * 2	35
Calendar	ε	Today is Monday, January 30, 2023.
Machine Translation	sûreté nucléaire	nuclear safety

Table 1: Examples of inputs and outputs for all APIs used.

API	Number of Examples $\tau_f = 0.5$	$\tau_f = 1.0$	$\tau_f = 2.0$
Question Answering	51,987	18,526	5,135
Wikipedia Search	207,241	60,974	13,944
Calculator	3,680	994	138
Calendar	61,811	20,587	3,007
Machine Translation	3,156	1,034	229

Table 2: Number of examples with API calls in C^* for different values of our filtering threshold τ_f.

Your task is to add calls to a Question Answering API to a piece of text. The questions should help you get information required to complete the text. You can call the API by writing "[QA(question)]" where "question" is the question you want to ask. Here are some examples of API calls:

Input: Joe Biden was born in Scranton, Pennsylvania.

Output: Joe Biden was born in [QA("Where was Joe Biden born?")] Scranton, [QA("In which state is Scranton?")] Pennsylvania.

Input: Coca-Cola, or Coke, is a carbonated soft drink manufactured by the Coca-Cola Company.

Output: Coca-Cola, or [QA("What other name is Coca-Cola known by?")] Coke, is a carbonated soft drink manufactured by [QA("Who manufactures Coca-Cola?")] the Coca-Cola Company.

Input: x

Output:

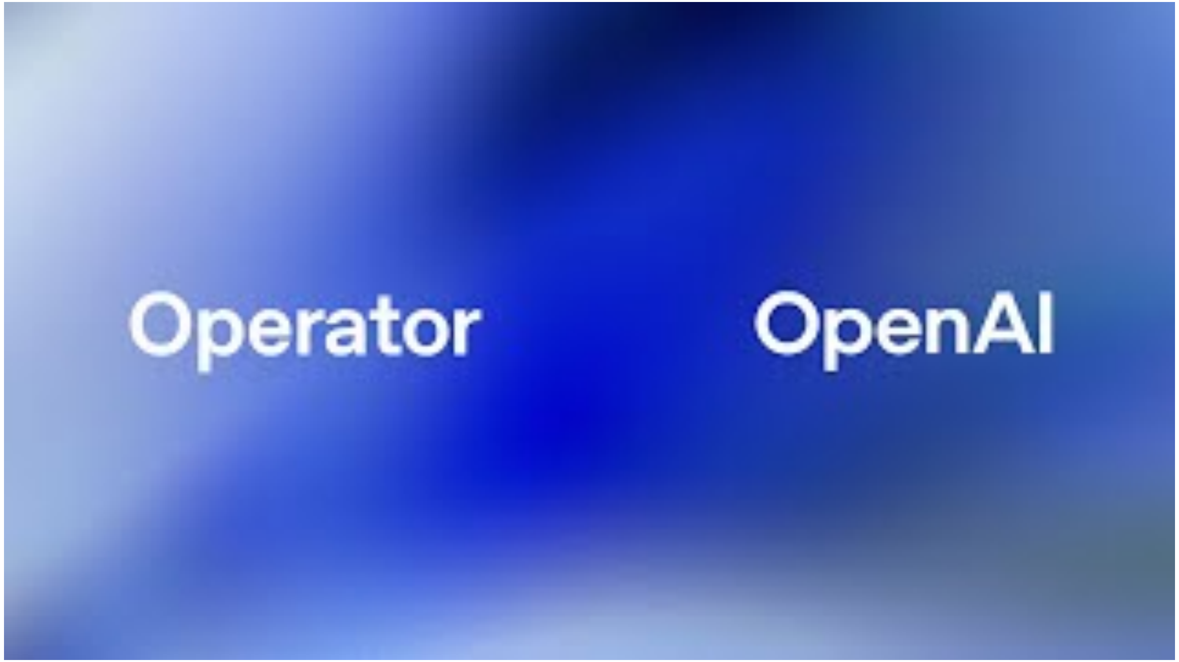
Figure 3: An exemplary prompt $P(\mathbf{x})$ used to generate API calls for the question answering tool.

AI Agent Applications (as Products)

- **OpenAI Operator (Web Agent)**
- **Deep Research (Research Agent)**
- **Copilot Agent (Coding agent)**
- **Manus (MCP + Research agent)**
- **Dify (open-source LLM app development platform with Agent capabilities)**

OpenAI Operator (Web Agent)

Query: *Find me a family friendly campsite at Joshua Tree this weekend*



Deep Research (Research Agent)

Query: *Compile a research report on how the retail industry has changed in the past 3 years. Use bullets and tablets where necessary for clarity*

Compile a research report on how the retail industry has changed in the past 3 years. Use bullets and tablets where necessary for clarity

To make sure the report meets your needs, could you please clarify a few things:

1. Are you interested in global retail trends or focused on a specific region (e.g., US, Europe, Asia)?
2. Should the report cover all types of retail (e.g., online, brick-and-mortar, hybrid), or focus on one?
3. Would you like emphasis on specific aspects like consumer behavior, technology, supply chain, labor, or financial performance?
4. Do you need data and trends from 2022 to 2025 specifically?

Once I have this info, I'll get started right away.

1) US, 2) online, 3) recommendation system, 4) could be flexible

Got it! I'll compile a report on how the U.S. online retail industry has evolved over the past three years, focusing specifically on changes and advancements in recommendation systems. I'll include trends, technology shifts, consumer behavior insights, and notable company innovations. I'll share the report with you as soon as it's ready.

Starting research

Got it! I'll compile a report on how the U.S. online retail industry has evolved over the past three years, focusing specifically on changes and advancements in recommendation systems. I'll include trends, technology shifts, consumer behavior insights, and notable company innovations. I'll share the report with you as soon as it's ready.

Research completed in 12m · 25 sources

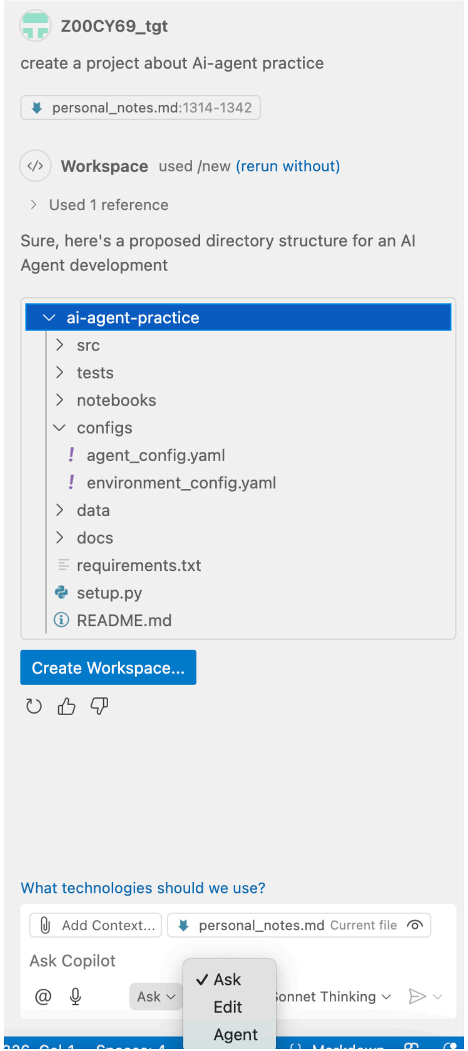
Evolution of Recommendation Systems in U.S. Online Retail (2022–2025)

The period from 2022 to 2025 has seen **significant transformations in the U.S. online retail industry's use of recommendation systems.** Once a novel feature, recommendation engines have become **AI-driven personalization powerhouses** central to e-commerce strategy. This report examines how technological advancements, key industry players, and shifting consumer expectations have reshaped online retail recommendations – all while balancing personalization with privacy. Below, we break down the major developments by category, with a focus on the U.S. market and trends.

Key Technological Advancements in Recommendation Systems (2022–2025)

Copilot Agent (Coding agent)

Query: *Create a project about Ai-agent practice*



Manus (MCP + Research agent)

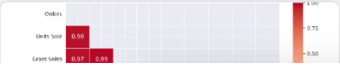
Use case gallery

Learn how Manus handles real-world tasks through step-by-step replays.

Featured | Research | Life | **Data Analysis** | Education | Productivity | WTF

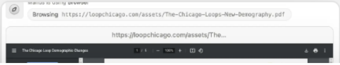
Online store operation analysis

Upload your Amazon store sales data and Manus delivers actionable insights, detailed visualizations, and customized strategies designed to increase your sales performance.



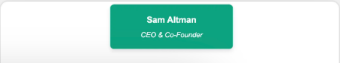
Formulation of store sales improvement strategies

Manus conducts thorough analysis of neighborhood demographics surrounding a Texas BBQ restaurant to develop targeted, data-backed strategies that effectively boost sales.

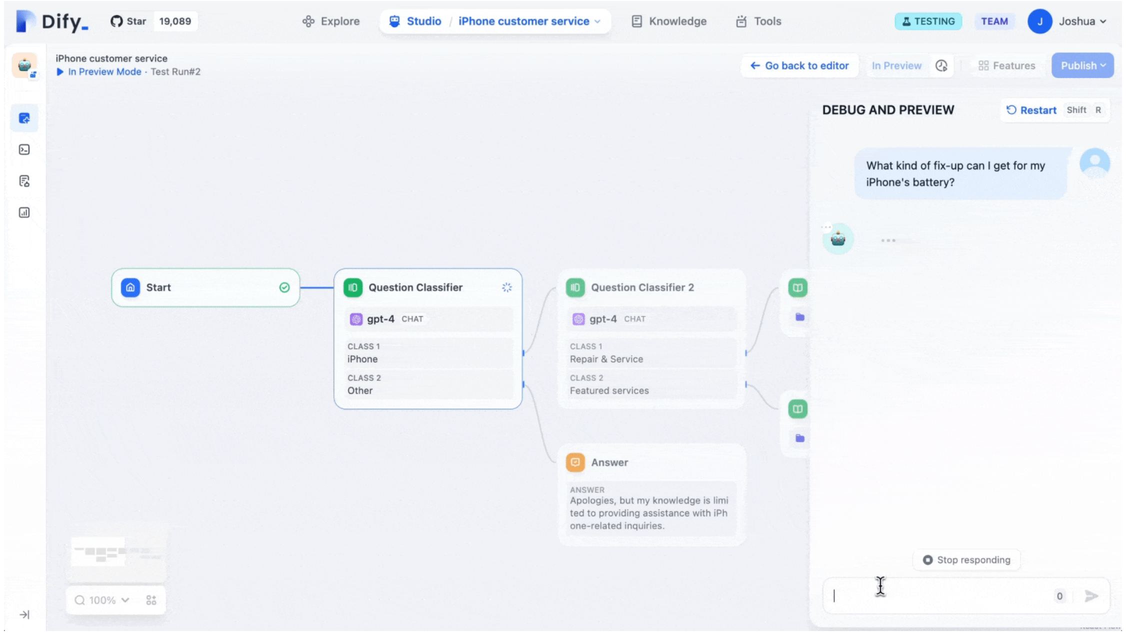


Create the company's organizational chart

Through extensive public information gathering and meticulous analysis, Manus creates precise and visually clear organizational charts tailored to your company's structure.



Dify: LLM framework



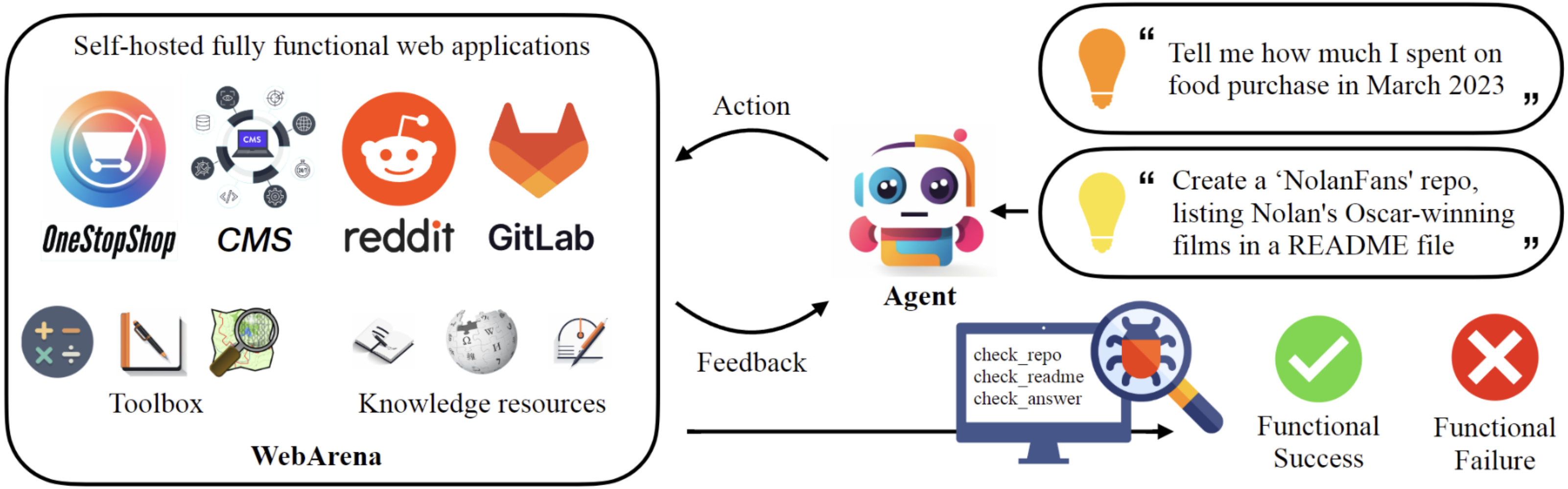
- Agent capabilities: You can define agents based on LLM Function Calling or ReAct, and add pre-built or custom tools for the agent. Dify provides 50+ built-in tools for AI agents, such as Google Search, DALL·E, Stable Diffusion and WolframAlpha.

Evaluation Benchmarks

- End-to-End Evaluations: Systems like **WebArena** and **GAIA** serve as testing grounds for the real-world performance of AI agents.

Criterion	WebArena	GAIA
Primary Focus	Web-based task execution and interaction	General assistant capabilities across diverse real-world questions
Evaluation Metrics	Functional correctness, task diversity, long-horizon planning, tool use	Reasoning ability, multi-modality handling, web browsing, tool use, task difficulty levels
Task Domains	E-commerce, forums, content management, collaborative development	Varied real-world scenarios requiring diverse skills
Benchmark Size	812 tasks	450 questions across three difficulty levels
Performance Benchmark	Success rate of task completion	Accuracy compared to human performance (~92%)

WebArena: A Realistic Web Environment for Building Autonomous Agents



🌐 WebArena Example Task

🔎 Task:

> *"Find a laptop under $1,000 with at least 16GB RAM and add it to the shopping cart."*

🧠 What the Agent Needs to Do:

1. Open the e-commerce website.
2. Use the **search bar** or **filters** to narrow down results.
3. Identify laptops matching both price and RAM criteria.
4. Click on a valid product.
5. Click **"Add to Cart"**.

✅ Evaluation: How It's Scored

✔ Success Criteria

WebArena uses **automated checkers** to validate:

- Did the agent **interact with the correct elements** (e.g. real search bar)?
- Was the **final state correct** (e.g. item in cart matching the constraints)?
- Was the **task completed within allowed steps/time**?

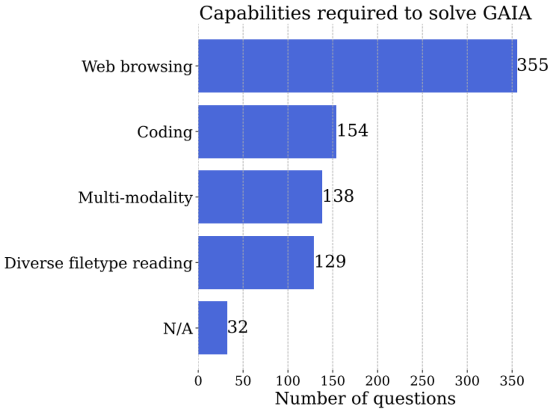
➡️ **Score**:

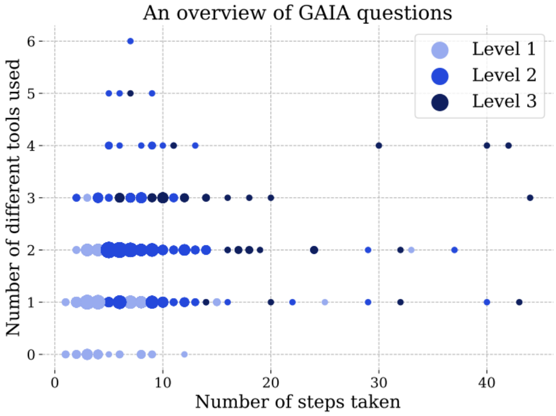
- **1.0** if all conditions met
- **0.0** if task failed (e.g. wrong item, didn't add to cart, ran out of steps)

There's no partial credit — it's **binary success** per task.

GAIA: A BENCHMARK FOR GENERAL AI ASSISTANTS

Grégoire Mialon[1], **Clémentine Fourrier**[2], **Craig Swift**[3], **Thomas Wolf** [2], **Yann LeCun**[1], **Thomas Scialom**[1]
[1]Meta AI, [2]Hugging Face, [3]AutoGPT
{gmialon, tscialom}@meta.com, clementine@huggingface.com





- Level 1: No tools or one tool; ≤ 5 steps
- Level 2: 5–10 steps; multiple tools required
- Level 3: Arbitrarily long sequences; many tools; general-world access

🧠 GAIA Task Levels with Examples

Level	Task Example	What Makes It This Level
Level 1 *(Simple)*	🟢 **"What is the capital of France?"**	- Requires basic recall or single search - No need for tools or reasoning - Completed in 1–2 steps
Level 2 *(Moderate)*	🟡 **"Compare the average rent in Boston and San Francisco, and suggest which is more affordable for students."**	- Requires 2+ tool uses (search rent data for both cities) - Reason about affordability - Involves ~5–10 steps
Level 3 *(Complex)*	🔴 **"Plan a 7-day Europe trip under $2,000, including flights, accommodations, and activities. Minimize carbon footprint."**	- Open-ended, multi-modal - Requires travel search, budget calculation, carbon impact comparison - 10+ steps, integrates many tools, planning & optimization logic

🧠 What is GAIA Testing?

GAIA is designed to test whether an AI assistant can:

- Reason through complex real-world tasks
- Use tools (like web search, calculators, APIs)
- Handle multi-step workflows
- Make factual, grounded decisions

Question:

"Find the cheapest direct round-trip flight from New York to Paris in June and summarize the best hotel options near the Eiffel Tower with high reviews."

🌀 What the AI Agent Needs to Do:

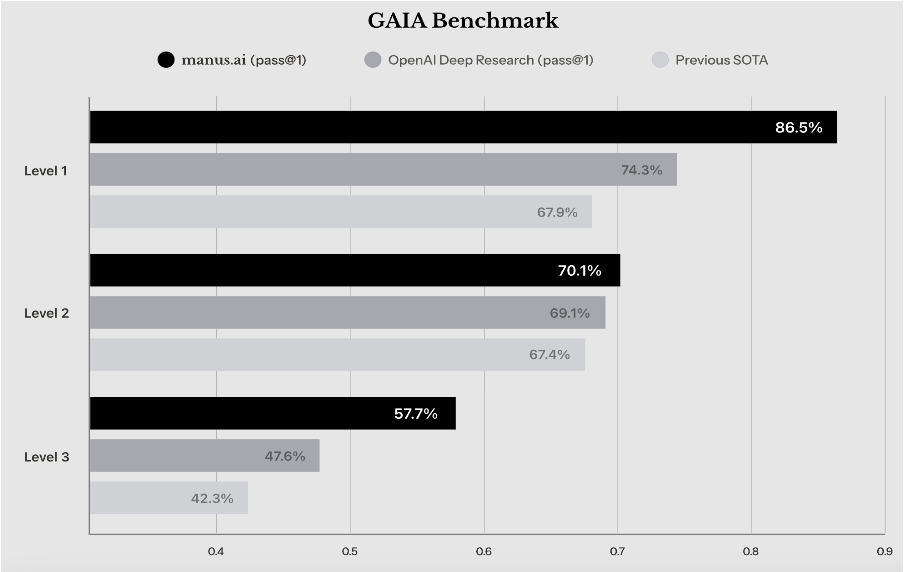
1. Understand and **decompose** the task (flight + hotel).
2. Use a **flight search tool or plugin** to:
   - Filter by "direct", "round-trip", "NY to Paris", "June".
3. Pick the **cheapest option** (e.g., $750 Air France).
4. Use another tool to search for **hotels near Eiffel Tower**:
   - Filter by review score (e.g., 4.5+).
   - Summarize 2–3 options with prices.
5. Return a clean, structured answer.

🧾 Sample Answer

> ✈️ **Flight:** Air France, round-trip, JFK → CDG, June 12–20, $749
> 🏨 **Hotel:**
> - "Hotel La Comtesse: 4.6★, 2 min walk from Eiffel, $210/night"
> - "Pullman Paris Tour Eiffel: 4.5★, direct view of Eiffel, $280/night
>   🔍 Based on reviews and proximity, La Comtesse is best for value."

GAIA

	Level 1	Level 2	Level 3	Avg.
Previous SOTA ↗	67.92	67.44	42.31	63.64
Deep Research (pass@1)	74.29	69.06	47.6	67.36
Deep Research (cons@64)	**78.66**	**73.21**	**58.03**	**72.57**



Trade-Off Considerations

- Evaluation benchmark: provide a comprehensive set of evaluation criteria to serve as a reference.
- Practical Factors: Cost, Latency, Scalability, Maintainability, Reliability, Security in real-world applications
- Benchmark Metrics: a composite "total score" to compare performance across different scenarios

Take Home Messages

- AI agents could be utilized for the business problems at Target
- From the advanced AI agent applications, we could leverage the tools for insights
- It is good to know how we can evaluate the AI agents considering the key components

Related Documents

- Language Agents: Foundations, Prospects, and Risks
- GAIA: A Benchmark for General AI Assistants
- WebArena: A Realistic Web Environment for Building Autonomous Agents
- ReAct: Synergizing Reasoning and Acting in Language Models
- ToolFormer: Language Models Can Teach Themselves to Use Tools
- Cognitive Architectures for Language Agents

Q & A